Wageningen Academic Publishers

Bovine mastitis may be associated with the deprivation of gut *Lactobacillus*

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Received: 9 April 2015 / Accepted: 25 August 2015

RESEARCH ARTICLE

Abstract

Bovine mastitis is an economical important microbial disease in dairy industry. Some recent human clinical trials have shown that oral probiotics supplementation could effectively control clinical mastitis, suggesting that the mechanism of mastitis protection might be achieved via the host gut microbiota. We aimed to test our hypothesis that bovine mastitis was related to changes in both the mammary and gut microbial profiles. By quantitative PCR, the milk and faecal microbial profiles of cows with low ($<3\times10^5$ cells/ml) and high ($>1\times10^6$ cells/ml) somatic cell count (SCC) were compared. Firstly, we observed drastic differences in both the milk and faecal microbial compositions at genus and *Lactobacillus*-species levels between the two groups. Secondly, the pattern of faecal microbial community changes of mastitis cows was similar to that of the milk, characterised by a general increase in the mastitis pathogens (*Enterococcus*, *Streptococcus* and *Staphylococcus*) and deprivation of *Lactobacillus* and its members (*L. salivarius*, *L. sakei*, *L. ruminis*, *L. delbrueckii*, *L. buchneri*, and *L. acidophilus*). Thirdly, only the faecal lactobacilli, but not bifidobacteria correlated with the milk microbial communities and SCC. Our data together hint to a close association between bovine mastitis, the host gut and milk microbiota.

Keywords: gastrointestinal tract, mammary infection, mastitis, gut microbiota, *Lactobacillus*, *Bifidobacterium*

1. Introduction

Bovine mastitis is an economical important disease in dairy industry. It is often caused by pathogens within the genera *Enterococcus*, *Staphylococcus*, *Streptococcus* and *Escherichia* (Akram *et al.*, 2013; Suojala *et al.*, 2011). Some of these pathogens produce toxins that can damage the mammary gland milk-producing tissue, and subsequently promote mammary inflammation. The inflammation causes decreased milk production and quality (Schroeder, 1997). Antibiotics are routinely used to treat bovine mastitis (Barlow, 2011), however, the concerns of spreading antibiotic resistance genes and low clinical treatment efficacy caused by multidrug-resistant strain infections prompt for alternative treatment approaches (Espeche *et al.*, 2012).

Recently, oral probiotics supplementation has been proposed as an alternative approach in combating mastitis (Fernández *et al.*, 2014) and optimistic therapeutic effects have been demonstrated in some human clinical trials. The milk-originated *Lactobacillus salivarius* CECT 5713 and *Lactobacillus gasseri* CECT 5714 significantly reduced the milk staphylococcal counts in women suffering from mastitis (Jiménez *et al.*, 2008). Similarly, in a large-scale study of 352 women with lactational mastitis, ingesting *Lactobacillus fermentum* CECT 5716 and *L. salivarius* CECT 5713 effectively diminished the milk staphylococcal and/or streptococcal counts to healthy levels. Moreover, these specific probiotic bacteria were detectable in the subjects' breast milks (Arroyo *et al.*, 2010). Thus, probiotic supplementation may be a potential anti-mastitis treatment in human.

The microbiology-related studies of bovine mastitis have mainly concentrated on profiling of mastitic milk microbiota (Oikonomou *et al.*, 2012; Verdier-Metz *et al.*, 2012). We previously showed that milk *Lactobacillus* was negatively correlated with bovine mastitis pathogens and severity (Qiao *et al.*, 2015). Although there is profound evidence in effective mastitis treatment by ingesting *Lactobacillus* and that a diminished milk *Lactobacillus* population may increase mastitis susceptibility in cow, this field remains understudied. Particularly, the knowledge of *Lactobacillus* profiles specific to healthy and diseased cows may shed light on the species that are protective against bovine mastitis.

The human clinical successes of controlling mastitis with oral probiotics have also opened interesting questions regarding how the gut microbiota are connected with responses at different host body locations, e.g. the mammary gland in the case of mastitis, and what the mechanisms of the observed therapeutic effects are. Presumably, the probiotic bacteria survived in the host gut upon ingestion, became part of the allochthonous gut microbiota, and exerted the anti-mastitic effect. Oral probiotics administration is able to temporally modulate the host gut microbiota in human subjects (Kwok *et al.*, 2014; Zhang *et al.*, 2014). The gut microbes and ingesting *Lactobacillus* can also improve obesity (Sanz *et al.*, 2013), immunity (Borchers *et al.*, 2009), mineral and bone metabolism (Parvaneh *et al.*, 2014), dental caries prevention (Cagetti *et al.*, 2013), modulate neural behaviour (Bravo *et al.*, 2011), etc., indicating that the gut microbiota influences a wide range of host body functions through, but not limited to, the gut.

Thus, this study hypothesises that both the mammary and gut microbiota are important in maintaining cattle udder health. We compared the milk and faecal microbes of low (healthy and subclinical mastitis cows) and high (clinical mastitis cows) somatic cell count (SCC) cows. We asked (1) whether and how the milk and faecal microbial communities of these two cow groups differed; and (2) whether SCC, the milk and the gut microbiota of especially the *Lactobacillus* and *Bifidobacterium* genera were correlated. Our findings have provided novel insights into the microbiology of bovine mastitis, particularly a possible linkage between the gut microbiota and bovine mastitis. Research in such aspect may open new perspectives for the management of dairy animals.

2. Materials and methods

Ethic statement

The study protocol was approved by the Ethical Committee of Inner Mongolia Agricultural University (Hohhot, China P.R.) and was permitted by the owners of the sampled dairy farm. Every effort was made to minimise animal suffering.

Sample collection and DNA extraction

Holstein cows of 3 to 6 years old were used in this study. Collection of faeces and milk samples were performed on the same day. The mastitis severity of cows was graded by the SCC, which is a commonly used parameter representing the leukocytes quantity per ml of milk. 72 individual quarter milk and faeces samples were collected from a dairy farm in Inner Mongolia, Hohhot: 36 from the faeces and milk samples of clinical mastitis cows (SCC $>1\times10^6$ cells/ml, mean ± standard error of the mean (SEM) $462.88\pm51.71\times10^4$ cells/ml; with redness and swelling around the udder tissue; Supplementary Table S1); while another 36 were obtained from healthy or subclinical mastitis cows (SCC $<3\times10^5$ cells/ml, mean ± SEM $10.66\pm1.32\times10^4$ cells/ml; with no clinical mastitis sign, e.g. abnormal milk production, redness and swelling around the udder tissue; Supplementary Table S1). The International Dairy Federation has recommended using an SCC of 200,000 cells/ml as cut-off value to define mastitis in composite bovine milk; however, it has recognised the challenge in setting a single bovine mastitis cut-off value and that the SCC range of subclinical mastitis cows widely varies (International Dairy Federation, 2013). Moreover, the SCC may vary subjected to individual cow specific factors (Sharma *et al.*, 2011). Thus, our study has chosen a slightly higher cut-off value of 300,000 cells/ml to distinguish healthy and subclinical mastitis cows from the severe mastitic ones ($>1\times10^6$ cells/ml). Aliquots of milk samples were sent to the laboratory immediately for SCC determination with a Bentley FTS/FCM400 Combi Instrument (Chaska, MN, USA). To prevent environmental microbial contamination, the cow udders and teats were wiped with cotton wool soaked in 70% alcohol and the first few streams of milk were discarded before sample collection.

Faeces and milk samples were collected aseptically and frozen immediately in liquid nitrogen before being transported to the laboratory. The total milk and faecal DNA were isolated by QIAamp DNA Food Kit and QIAamp Stool Mini Kit (QIAgen, Hilden, Germany), respectively. The milk samples (20 ml) were centrifuged at 7,150×*g* for 20 min, and the supernatant was removed. Then, total DNA was isolated from the pellets using a modified protocol of the QIAamp DNA Food Kit. Faecal DNA was extracted from 0.4 g of sample using a modified protocol of the QIAamp Stool Mini Kit. DNA was eluted in ddH_2O and stored at -20 °C until use.

Quantitative PCR

The PCR primers are shown in Supplementary Table S2. qPCR was performed using a Step-One™ Real-Time PCR System (Applied Biosystems, Foster City, CA, USA). Reactions were performed in 96-well plates with SYBR® Premix Ex TaqTM II (TaKaRa Bio, Otsu, Japan). All PCR reactions were done in triplicate with a reaction volume of 20 µl with 0.4 µM final concentration of each primer, a fixed genomic DNA amount (100 ng) and an appropriate amount of ddH_2O. The amplification program was 1 cycle at 95 °C for 3 min before 40 cycles at 95 °C for 30 s, 62 °C for 40 s, and 72 °C for 1 min. The fluorescent product was detected at the last step of each cycle.

Standard curves for each qPCR assay were generated by plotting the threshold cycle (CT) values against target copy numbers corresponding to serially diluted plasmid standards (Integrated DNA Technologies, Coralville, IA, USA). Each qPCR assay was calibrated with the specific plasmid construct containing the respective target gene fragment. The target copy numbers (T) were estimated by the equation $T=[D/(PL\times660)]\times6.022\times10^{23}$, where D (g/l) and PL (in base pairs) were the plasmid DNA concentration and length, respectively. Each standard curve was generated from at least five 10-fold plasmid dilutions in triplicate. Data generated by qPCR were expressed in log gene copy number per gram or ml of sample.

Statistical analyses

Data were expressed in mean ± SEM. Significant differences between groups were assessed with the Mann-Whitney test. Non-metric multidimensional scaling (n-mds), analysis of similarity (ANOSIM) (permutation with 9,999 replicates) and similarity percentage (SIMPER) (all based on Bray-Curtis distance) were used to evaluate the microbial structural differences. All statistical analyses were performed with the PAST software (Hammer *et al.*, 2001).

3. Results and discussion

Faecal and milk microbial communities of low and high SCC cows

To test our hypothesis that both the mammary and gut microbiota are linked to the cattle udder health status, we first compared the quantities of the total bacteria, anaerobic fungi, *Bacteroides*, *Escherichia coli*, *Enterococcus*, *Staphylococcus*, *Streptococcus*, *Bifidobacterium*, *Lactobacillus*, *Lactococcus lactis* in the milk and faeces of the low and high SCC cows (each n=36) by qPCR (Figure 1, Supplementary Table S3).

Our results detected significant differences in the milk from the low and high SCC cows for most of the quantified microbes (P<0.0347), except for *Staphylococcus* and *L. gasseri* (P=0.1298 and 0.1712, respectively). More potential mastitis pathogens were found in the milk of the high SCC group (P<0.0101 for *E. coli*, *Enterococcus*, and *Streptococcus*) (Supplementary Table S3). Meanwhile, fewer milk *Lactobacillus* at both genus (P=0.0347) and species (11 out of 12 species, except for *L. gasseri*; P<0.0007) levels were observed in the high SCC group (Figure 2, Supplementary Table S3).

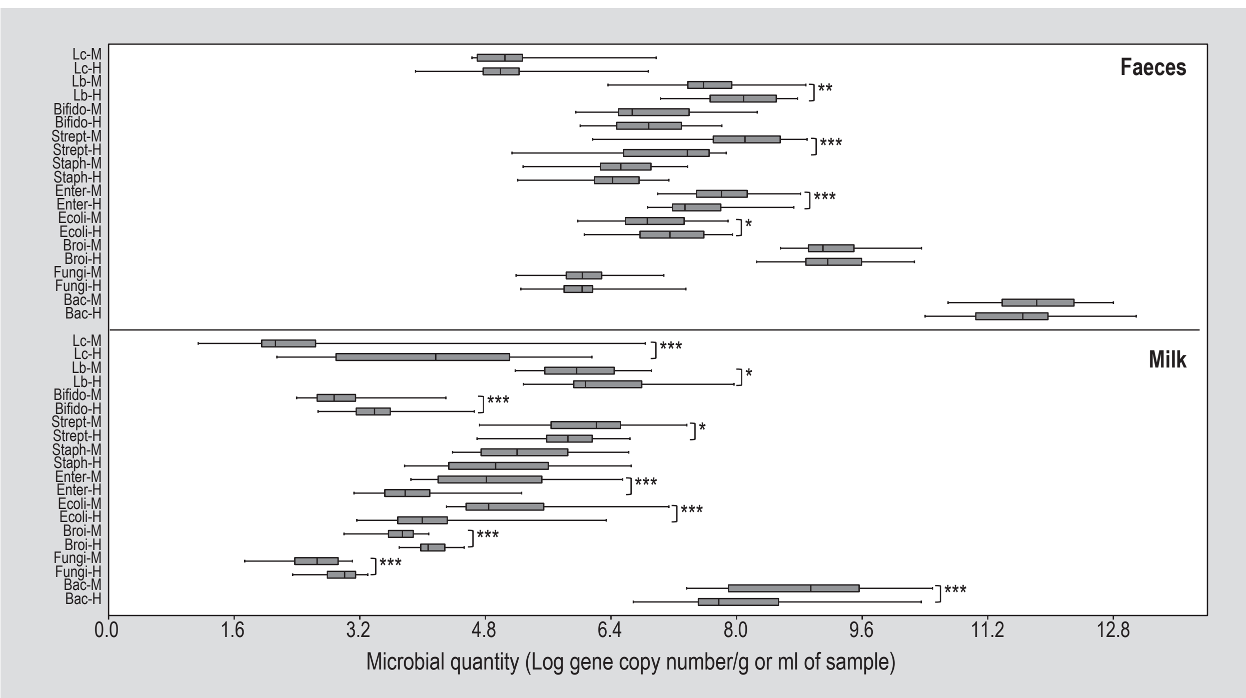

Figure 1. Quantification of faecal and milk microbes in the two groups of cows. Significantly difference between sample pairs was assessed by pairwise Mann-Whitney test. Significant values are represented by * (*P*<0.001), ** (*P*<0.01), and * (*P*<0.05), respectively. Lc = *Lactococcus lactis*; Lb = *Lactobacillus*; Bifido = *Bifidobacterium*; Strept = *Streptococcus*; Staph = *Staphylococcus*; Enter = *Enterococcus*; Ecoli = *Escherichia coli*; Broi = *Bacteroides*; Bac = total bacteria. H and M represent cows with low (<300,000 cells/ml) and high (>1 million cells/ml) somatic cell count (SCC).**

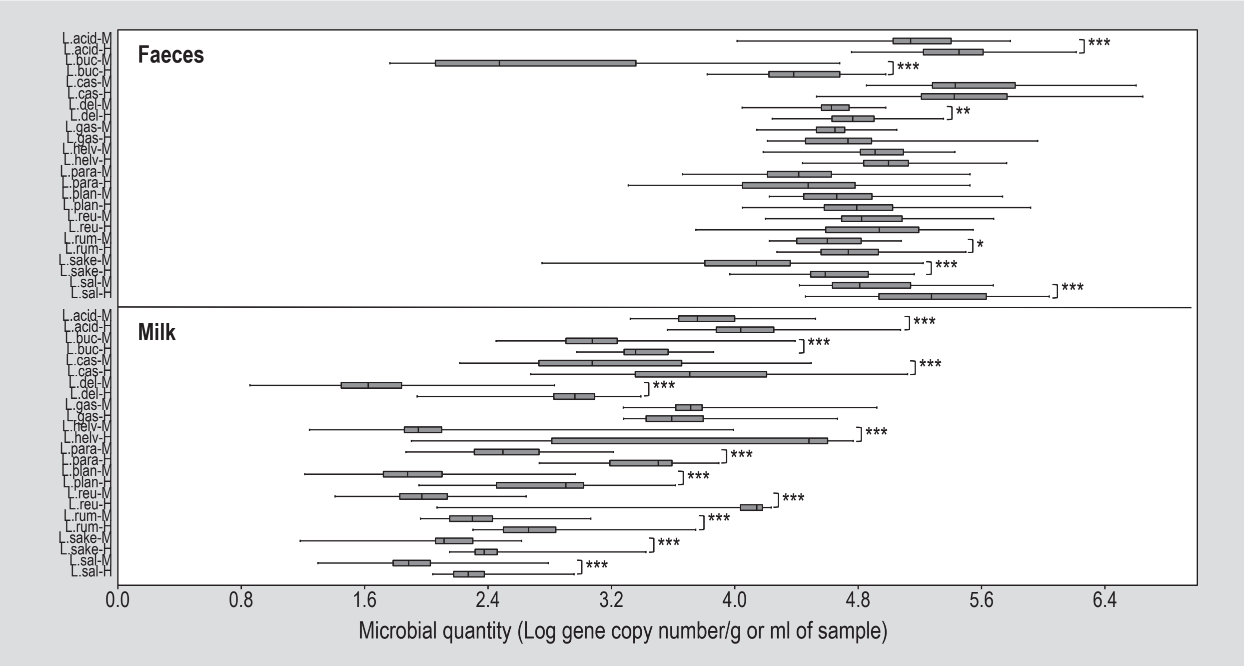

Figure 2. Quantification of faecal and milk *Lactobacillus* species in the two groups of cows. Significantly difference between sample pairs was assessed by pairwise Mann-Whitney test. Significant values are represented by * (*P*<0.001), ** (*P*<0.01), and * (*P*<0.05), respectively. L.acid = *L. acidophilus*; L.buc = *L. buchneri*; L.cas = *L. casei*; L.del = *L. delbrueckii*; L.gas = *L. gasseri*; L.helv = *L. helveticus*; L.para = *L. paracasei*; L.plan = *L. plantarum*; L.reu = *L. reuteri*; L.rum = *L. ruminis*; L.sake = *L. sakei*; L.sal = *L. salivarius*. H and M represent cows with low (<300,000 cells/ml) and high (>1 million cells/ml) somatic cell count (SCC).**

In contrast, no significant differences existed in the total faecal bacterial and fungal amounts of the two sample groups (P=0.1162 and 0.6728, respectively). Significant differences between the faecal samples of the two groups were found in *E. coli*, *Enterococcus*, *Streptococcus*, *Lactobacillus*, *L. salivarius*, *Lactobacillus sakei*, *Lactobacillus ruminis*, *Lactobacillus delbrueckii*, *Lactobacillus buchneri*, and *Lactobacillus acidophilus* (P=0.0000-0.0432). Similar to the results from milk, the genera *Enterococcus* and *Streptococcus* were higher in the faeces of the high SCC group (P=0.0005 and P=0.0000, respectively), whereas the genus *Lactobacillus* (P=0.0024) and most *Lactobacillus* species generally decreased (Figure 1, Figure 2, Supplementary Table S3).

Non-metric multidimensional scaling (n-mds) is a robust statistical technique that creates a score plot displaying the relative positions of a number of objects, indicating the perceived differences between the microbial profile of milk and faeces samples in our case. Thus, in order to visualise the overall difference of the milk and faecal bacteria profiles, the data were evaluated by n-mds. Symbols representing the faecal and milk samples were located separately on the plot (Figure 3), suggesting that the overall bacterial profile of the faeces regardless of the animal health status was distinct from that of the milk. However, between the milk or faecal samples of the low and high SCC groups, no distinct clustering pattern could be identified, indicating that the overall bacterial profile of the milk or the faecal samples was highly similar irrespective to the animal health status. The high similarity within subgroups was supported by the low R-values (0.0280 and 0.0853 for the faeces and milk, respectively) by ANOSIM.

The elevation of pathogens in the mastitis milk microbiota has previously been reported (Bhatt *et al.*, 2012; Oikonomou *et al.*, 2012). Mastitis is often related to bacterial infection of the mammary gland, subsequently resulting in mammary alveolus damage, inflammatory cell infiltration, mammary gland oedema and interstitial haemorrhage (Li *et al.*, 2013). The infiltration of inflammatory cells leads to a high SCC that fights locally against pathogens, thus altering the milk microbiota. Moreover, some mastitis pathogens can secrete antibacterial substances like bacteriocins, organic acids and hydrogen peroxide (Bogni *et al.*, 2011; Espeche *et al.*, 2012) that further contribute to changing the local microbiota. Our data surprisingly revealed a similar pattern of structural faecal microbial change as compared to the milk in mastitis animals, which was common in their lactobacilli deprivation, although at this point it is difficult to conclude if the reduction of lactobacilli directly contributed to mastitis protection or if it was just a consequence of the infection itself. Overall speaking, our data nevertheless documented an alteration of both the mammary and gut microbial composition, particularly of the *Lactobacillus* genus, in the case of bovine mastitis.

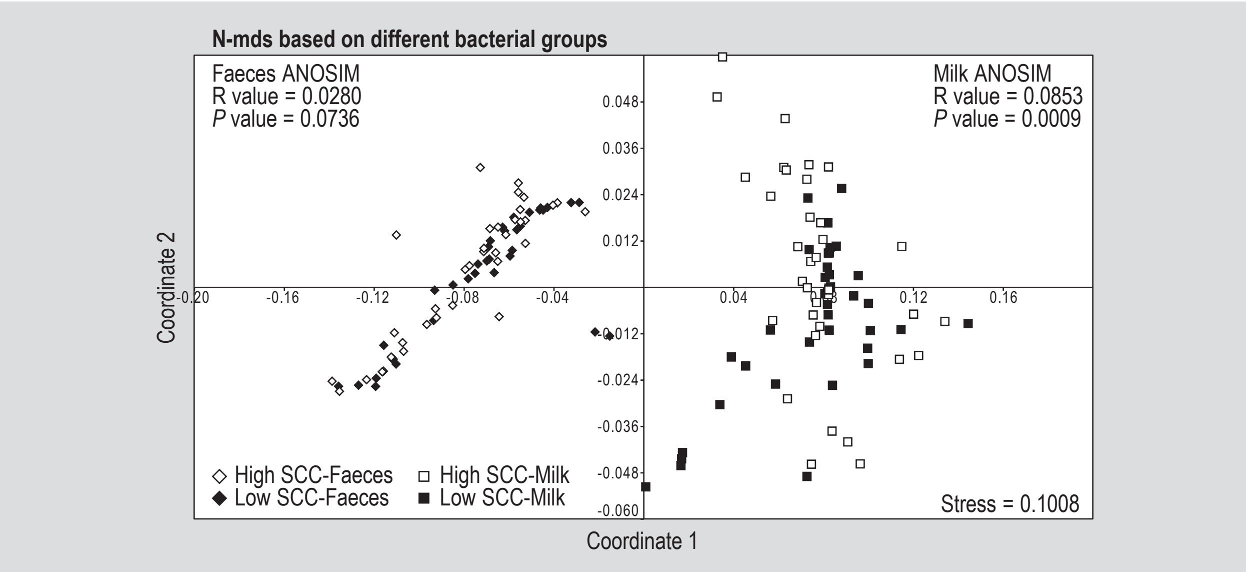

Figure 3. Non-metric multidimensional scale (n-mds) plot and analysis of similarity (ANOSIM) based on different bacterial groups. A low stress value of 0.1008 was obtained, indicating that the dataset was well-represented by the imposed analysis model. The ANOSIM R-value scales from +1 and -1. A high R-value indicates that all the most similar samples are within the same groups and vice versa, while R=0 suggests that the high and low similarities are perfectly mixed and bear no relationship to the group. Any significant difference between different groups is represented by the *P*-value. SCC = somatic cell count.

Correlation between the faecal *Lactobacillus* and *Bifidobacterium* with SCC and milk mastitis pathogens

To determine whether the two commonly recognised gut probiotic genera, *Lactobacillus* and *Bifidobacterium*, were related to mastitis status, Spearman correlation analysis was performed to identify if any correlation existed between these two genera, with SCC and the milk microflora (Table 1). Interestingly, the faecal lactobacilli showed negative correlation with both SCC (r=-0.5331, *P*=0.0000) and the potential mastitis pathogens in the milk (*E. coli*, *Enterococcus*, *Staphylococcus* and *Streptococcus*; r=-0.1967 to -0.4765, *P*<0.0977). Moreover, the faecal lactobacilli were positively correlated with the beneficial milk bacteria (*Bifidobacterium*, *Lactobacillus* and *L. lactis*; r=0.2707 to 0.4163, *P*=0.0003 to 0.0215). In contrast, the faecal *Bifidobacterium* did not display any significant correlation with any of the milk microbes and SCC (*P*>0.05 in all cases) (Table 1). Such results strongly suggest that the faecal *Lactobacillus* but not *Bifidobacterium* were possibly involved in regulating the host milk microbiota.

Table 1. Spearman correlation between faecal *Lactobacillus* and *Bifidobacterium* with somatic cell count (SCC) and milk microbes.

	Faecal *Lactobacillus*		Faecal *Bifidobacterium*	
	r	*P*-value	r	*P*-value
SCC	-0.5331	0.0000	-0.0777	0.5165
Staphylococcus	-0.2561	0.0299	0.0605	0.6139
Enterococcus	-0.4130	0.0003	-0.0725	0.5450
Escherichia coli	-0.4765	0.0000	0.0214	0.8585
Streptococcus	-0.1967	0.0977	-0.1040	0.3847
Bifidobacterium	0.4163	0.0003	0.0848	0.4790
Lactobacillus	0.2707	0.0215	-0.0354	0.7680
Lactococcus lactis	0.3993	0.0005	0.0665	0.5790

Lactobacillus profiles of low and high SCC cows

Having identified the deprivation of gut lactobacilli in mastitis cows and its significant negative correlation with the potential pathogens, we further evaluated the *Lactobacillus*-profiles of the two cow groups by n-mds. Symbols representing the faecal and milk samples were located separately on the plot (Figure 4), indicating a distinct *Lactobacillus* composition of the faecal and milk samples. ANOSIM analysis on the faecal and milk samples returned R-values respectively of 0.1244 (*P*=0.0004) and 0.7205 (*P*=0.0001), suggesting that structural differences existed in both the milk and faecal samples between healthy and mastitis cows, with a greater difference in the milk.

The SIMPER procedure calculated the contribution of each *Lactobacillus* species in the structural dissimilarity (Table 2). For the milk, *Lactobacillus reuteri* ranked the highest contributing to 19.46% of the dissimilarity, followed by *Lactobacillus helveticus* (19.34%) and *L. delbrueckii* (12.24%). The other 9 *Lactobacillus* species

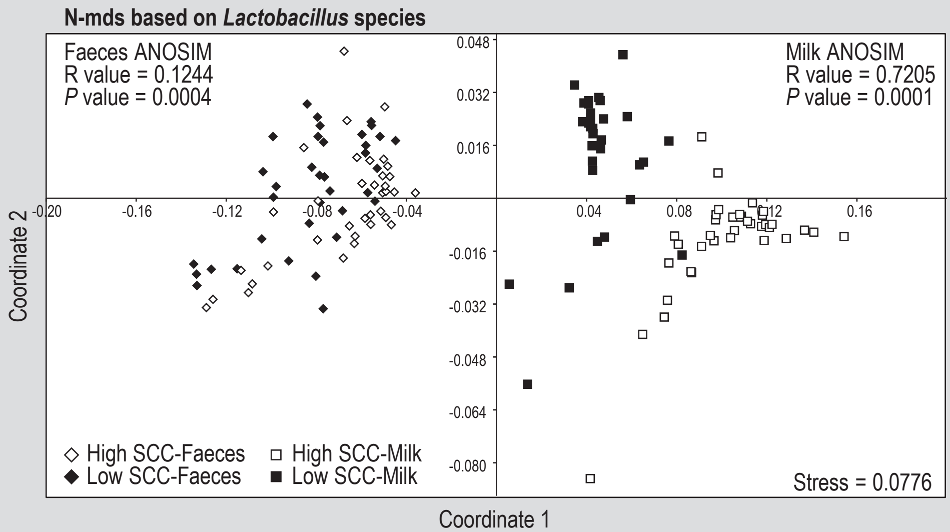

Figure 4. Non-metric multidimensional scale (n-mds) plot and analysis of similarity (ANOSIM) based on 12 *Lactobacillus* species. A low stress value of 0.0776 was obtained, indicating that the dataset was well-represented by the imposed analysis model. The ANOSIM R-value scales from +1 and -1. A high R-value indicates that all the most similar samples are within the same groups and vice versa, while R=0 suggests that the high and low similarities are perfectly mixed and bear no relationship to the group. Any significant difference between different groups is represented by the *P*-value. SCC = somatic cell count.

Table 2. Contribution of dissimilarity by faecal and milk *Lactobacillus* species with similarity percentage (SIMPER) analysis.

Milk			Faeces		
***Lactobacillus* species**	**Average dissimilarity (Bray-Curtis distance)**	**Contribution of dissimilarity (%)**	***Lactobacillus* species**	**Average dissimilarity (Bray-Curtis distance)**	**Contribution of dissimilarity (%)**
L. reuteri	2.8290	19.46	*L. buchneri*	1.5060	27.32
L. helveticus	2.8120	19.34	*L. sake*	0.5311	9.64
L. delbrueckii	1.7790	12.24	*L. salivarius*	0.4701	8.53
L. plantarum	1.3160	9.05	*L. casei*	0.4513	8.19
L. paracasei	1.2790	8.80	*L. paracasei*	0.4241	7.70
L. casei	1.1540	7.94	*L. plantarum*	0.3885	7.05
L. ruminis	0.6538	4.50	*L. acidophilus*	0.3674	6.67
L. salivarius	0.6501	4.47	*L. reuteri*	0.3467	6.29
L. buchneri	0.6061	4.17	*L. ruminis*	0.2663	4.83
L. acidophilus	0.5377	3.70	*L. helveticus*	0.2596	4.71
L. sake	0.5303	3.65	*L. gasseri*	0.2554	4.63
L. gasseri	0.3883	2.67	*L. delbrueckii*	0.2452	4.45

each contributed to less than 10% of the dissimilarity. The major contributor for the dissimilarity in the faeces was *L. buchneri* (contributing to 27.32% of the variation), while other species each contributed to less than 10% of the difference. This species also displayed the strongest negative correlation with SCC (r=-0.5449, *P*=0.0000), followed by *L. sakei* (r=-0.4859, *P*=0.0000) and *L. salivarius* (r=-0.3697, *P*=0.0014) (Table S4). The latter species only contributed respectively to 9.46%, and 8.53% of the dissimilarity of the faecal *Lactobacillus* profile between the low and high SCC groups (Table 2).

Several *Lactobacillus* species were found to bear a negative correlation with SCC. However, the species *L. buchneri* has so far received the most attention in the field of dairy mastitis, as it is considered by the European Food Safety Authority as a safe technological additive for improving the ensiling process for cattle and sheep (EFSA, 2014). Although it is still unclear about the role *L. buchneri* in bovine mastitis protection, feeding lactating cows with alfafa silage treated with *L. buchneri* and glycoside hydrolases could increase milk production (Kung *et al.*, 2003). Moreover, *L. buchneri* possesses an interesting metabolic pathway that converts lactate to acetate; acetate exhibits antimicrobial activities and enhances silage stability (Holzer *et al.*, 2003). Such antimicrobial activity may modulate the gut microbiota after silage ingestion. Thus, it may be worth further confirming the biological significance of the phenomenon of *L. buchneri* deprivation.

The species *L. acidophilus*, *L. salivarius* and *L. sakei* also displayed moderately strong negative correlation with SCC. Members of *L. salivarius* (Arroyo *et al.*, 2010; Jiménez *et al.*, 2008) have been reported to be effective in treating human mastitis. The strain *L. acidophilus* A was antagonistic to the growth of the mastitis pathogen *E. coli* in milk (Fang *et al.*, 1996), though the mechanism of such activity was not deciphered. To our knowledge, no direct study has been performed to explore the role of *L. sakei* in bovine mastitis. Noticeably, members of all three *Lactobacillus* species are known to produce bacteriocins, which may potentially play an active role in controlling mastitis pathogens *in situ* and account for their negative correlation with SCC, i.e. one of the indicators for host mastitis. On the other hand, the increased level of SCC in mastitic animals may involve in combating bacteria, including *Lactobacillus*, that exist at the mammary tissue. An inhibition of *L. acidophilus* by the elevated phagocytic activity of polymorphonuclear leucocytes was observed during the fermentation of somatic cell-containing milk (Okello-Uma and Marshall, 1986).

Accumulating evidence has unarguably shown the significance of gut microbiota in host health maintenance, but has seldom demonstrated its influence in mastitis. The current study reports several interesting findings: (1) different microbial milk and faecal profiles existed between relatively healthy and severe mastitis cows, which were characterised by the deprivation of lactobacilli in the latter group; (2) only the gut lactobacilli, but not bifidobacteria negatively correlated with SCC and the milk mastitis pathogens, suggesting that if there was modulation of mastitis by gut microflora, it would have been microbe-specific; (3) potentially, prominent reduction of the faecal *L. buchneri* occurred in cows suffering from severe mastitis. The biological significance of this species in bovine mastitis control is worth further exploring. Our study has revealed a potential interesting link between some members of the gut and mammary microbes, which merits further characterisation, in particular, how they ultimately influence the host health status, e.g. in the case of bovine mastitis. However, our observations are only preliminary, and our current findings are limited by the current method of study. We have chosen the qPCR approach for microbial enumeration because of its low cost, simplicity and rapidity in achieving a rough profile. Thus, it is a good method for an initial estimation of target microbial quantities. Yet, qPCR may over or under estimate the microbial quantities, depending on the concentration and copy number of target microbes in the samples, as well as the choice of standard strains. Moreover, our study provides no information on the viability and functionality of the detected microbes. Certainly, further studies should take advantage of the more sophisticated and integrated 'omics' approach, e.g. metagenomics, transcriptomics together with metabolomics, which will likely provide a more comprehensive microbial profiles in bovine mastitis. Such studies may also give insights into the mechanisms of how microbes interact with the host health. Nevertheless, we have identified specific *Lactobacillus* species that deserve future testing for their efficacy in controlling bovine mastitis. The identification of strains that can effectively combat bovine mastitis will improve the current practice of mastitis prevention and control, potentially reducing or even replacing the traditional antibiotic therapy, hence, may open new perspectives for the management of dairy animals.

Supplementary material

Supplementary material can be found online at http://dx.doi.org/10.3920/BM2015.0048.

Table S1. Somatic cell count in the milk samples.

Table S2. Oligonucleotide primers used in this study.

Table S3. Differences in bacterial quantities in the faecal and milk samples of healthy and mastitis cows.

Table S4. Spearman correlation between individual faecal *Lactobacillus* species and somatic cell count.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (Grant No. 31460389) and the China Agriculture Research System (Grant No. CARS-37).

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